

Class 12 Physics - Advanced Concepts

Electrostatics

Coulomb's Law

Force between two point charges:

$$F = k(q_1 q_2)/r^2$$

Where $k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$

Electric Field

$$E = F/q = kQ/r^2$$

Direction: Away from positive charge, towards negative charge

Gauss's Law

Electric flux through a closed surface equals charge enclosed divided by ϵ_0 :

$$\Phi = Q/\epsilon_0$$

Capacitance

$$C = Q/V$$

For parallel plate capacitor: $C = \epsilon_0 A/d$

Energy stored: $U = 1/2 CV^2$

Current Electricity

Kirchhoff's Laws

Kirchhoff's Current Law (KCL)

Sum of currents entering a junction = Sum of currents leaving the junction

Kirchhoff's Voltage Law (KVL)

In any closed loop, the sum of EMFs equals the sum of voltage drops

Wheatstone Bridge

Balanced condition: $P/Q = R/S$

Meter Bridge

Used to find unknown resistance using balanced Wheatstone bridge principle

Magnetic Effects of Current and Magnetism

Biot-Savart Law

$$dB = (\mu_0/4\pi) \times (I dl \sin \theta)/r^2$$

Ampere's Circuital Law

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

Magnetic Force

$F = qvB \sin \theta$ (on moving charge)

$F = BIL \sin \theta$ (on current-carrying conductor)

Magnetic Dipole Moment

$$M = NIA$$

Where N = number of turns, I = current, A = area

Electromagnetic Induction

Faraday's Laws

1. Change in magnetic flux induces EMF
2. Induced EMF = $-d\Phi/dt$

Lenz's Law

Direction of induced current opposes the change causing it

Self and Mutual Inductance

Self inductance: $\mathcal{E} = -L(di/dt)$

Mutual inductance: $\mathcal{E}_2 = -M(di_1/dt)$

Alternating Current

AC Voltage and Current

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t \pm \phi)$$

RMS Values

$$V_{\text{RMS}} = V_m / \sqrt{2}$$

$$I_{\text{RMS}} = I_m / \sqrt{2}$$

Impedance

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Where $X_L = \omega L$, $X_C = 1/\omega C$

Power Factor

$$\cos \phi = R/Z$$

$$\text{Average power: } P = V_{\text{RMS}} I_{\text{RMS}} \cos \phi$$

Electromagnetic Waves

$$\text{Speed of light: } c = 1/\sqrt{\mu_0 \epsilon_0} = 3 \times 10^8 \text{ m/s}$$

Electromagnetic Spectrum (increasing wavelength):

Gamma rays $\rightarrow$ X-rays $\rightarrow$ UV $\rightarrow$ Visible $\rightarrow$ IR $\rightarrow$ Microwaves $\rightarrow$ Radio waves

Ray Optics

Lens Formula

$$1/f = 1/v - 1/u$$

Lens Maker's Formula

$$1/f = (\mu - 1)(1/R_1 - 1/R_2)$$

Power of Lens

$$P = 1/f \text{ (in diopters when } f \text{ is in meters)}$$

Magnification

$$m = v/u = h'/h$$

Wave Optics

Young's Double Slit Experiment

Fringe width: $\beta = \lambda D/d$

Where λ = wavelength, D = distance to screen, d = slit separation

Interference Conditions

Constructive: Path difference = $n\lambda$

Destructive: Path difference = $(n + 1/2)\lambda$

Modern Physics

Photoelectric Effect

Einstein's equation: $h\nu = \phi + KE_{\text{max}}$

Where h = Planck's constant, ν = frequency, ϕ = work function

de Broglie Wavelength

$\lambda = h/p = h/mv$

Bohr's Atomic Model

Energy of electron: $E_n = -13.6/n^2 \text{ eV}$

Radius of orbit: $r_n = n^2 r_0$

Radioactivity

$N = N_0 e^{(-\lambda t)}$

Half-life: $t_{1/2} = 0.693/\lambda$

Important Constants

- Speed of light: $c = 3 \times 10^8 \text{ m/s}$
- Planck's constant: $h = 6.626 \times 10^{-34} \text{ Js}$
- Electron charge: $e = 1.6 \times 10^{-19} \text{ C}$
- Permittivity: $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
- Permeability: $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$

Practice numerical problems daily and understand the derivations!